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Generalising the configurations of an $N \times N \times N$ Rubik's Cube

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1 Introduction and methodology

Ernő Rubik invented the Rubik's Cube in 1974 [12]. It was the first twisty puzzle of its time and has maintained its spot as one of the world's most popular toys. Péter Sebestény invented the Rubik's Revenge (the $4 \times 4 \times 4$ cube) in 1981, naming it as a follow-up to the Rubik's Cube [14]. Since the age of 9, twisty puzzles like these have captivated my thoughts through the sport of *speedcubing*. Through writing this article, I intended to improve my mathematical understanding of these cubes.

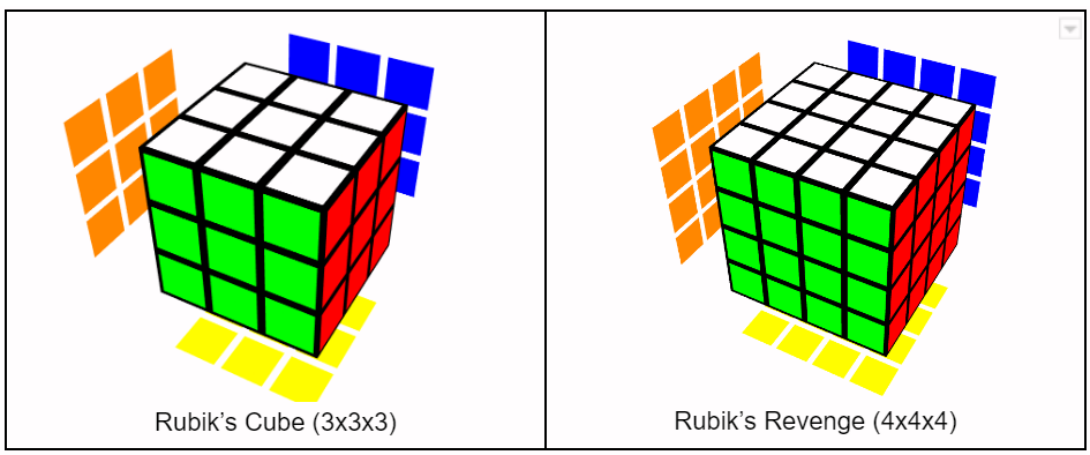

Figure 1: $3 \times 3 \times 3$ *and* $4 \times 4 \times 4$ *cubes*

Both cubes have six uniquely coloured faces, with red being opposite orange, green being opposite blue, and white being opposite yellow in their solved states. Each face can be rotated 90° in a clockwise or counter-clockwise direction. The cubes are solved when all six sides have solid colours.

The $3 \times 3 \times 3$ cube has a fixed centre on each side, meaning that, regardless of scramble, the relative position of each centre always stays the same. On the other hand, the $4 \times 4 \times 4$ cube does not have fixed centres, meaning that, despite its solved state still having the same colour relativity as the $3 \times 3 \times 3$ cube, there is no fixed piece like a centre that defines the colour relativity when the cube is scrambled.

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This introduces a concept called parity. Mathematically, parity is an integer's property of whether it is odd or even [16]. Regarding the cube, parity dictates whether the number of pieces that must be swapped is odd or even. As will be elucidated in this article, odd-layered cubes can only have even parity, while even-layered cubes can have even or odd parity, which affects both puzzles' total possible number of configurations.

It is essential to recognize the physical constraints of the puzzles. As both cubes have limited movement, each has a certain number of possible configurations. I will first investigate an odd ($3 \times 3 \times 3$) and even-layered cube's ($4 \times 4 \times 4$) total number of configurations to understand patterns in their respective calculations. Then, using these patterns, I will create a method to deduce an $N \times N \times N$ cube's total number of configurations.

2 Notation

The $3 \times 3 \times 3$ cube has three different types of pieces: 6 centres, 12 edges, and 8 corners:

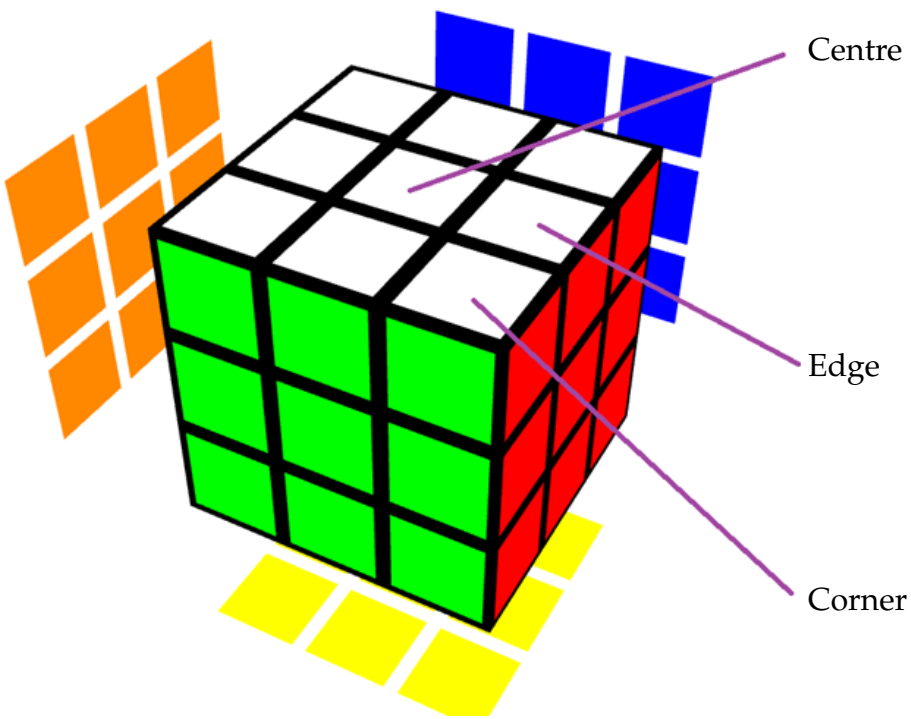


Figure 2: Types of $3 \times 3 \times 3$ cube pieces

Six different turns can be performed with these pieces:

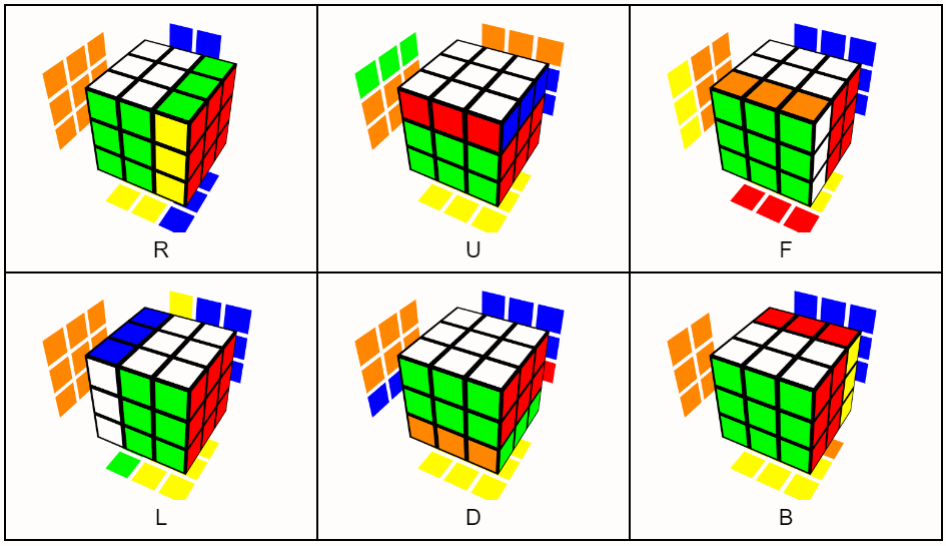


Figure 3: All possible $3 \times 3 \times 3$ moves

All these are 90° clockwise turns but also represent the cube's faces, with R representing the right face, L the left face, U the up face, D the down face, F the front face, and B the back face.

generalising, any 90° clockwise turn Y can be applied three times before arriving back at the original position. The number of applications of Y can be described in the superscript notation: Y, Y^2, Y^3. Any more applications of this Y turn can be described using these three notations, including counter-clockwise turns that can be characterized using negative superscripts [3].

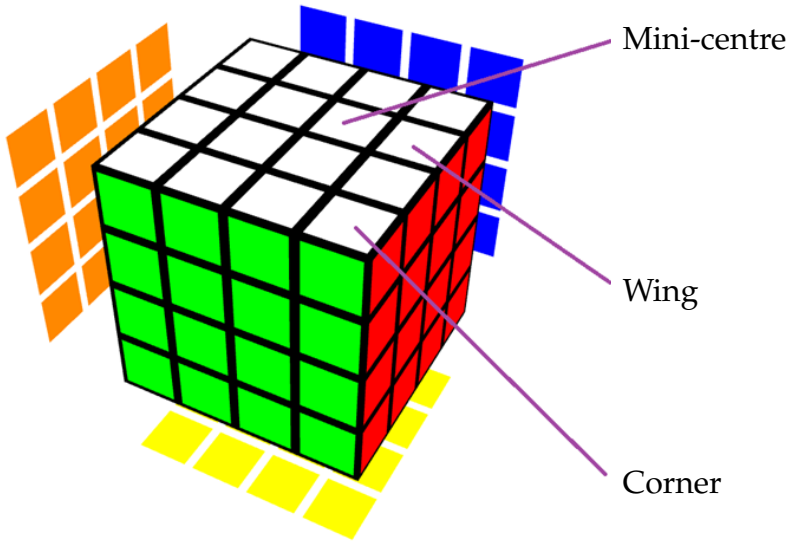


Figure 4: Types of $4 \times 4 \times 4$ cube pieces

The Rubik's Revenge works similarly, but with slight differences. It also has three different types of pieces: 24 mini-centres, 24 wings, and 8 corners.

12 different turns can be performed with these pieces:

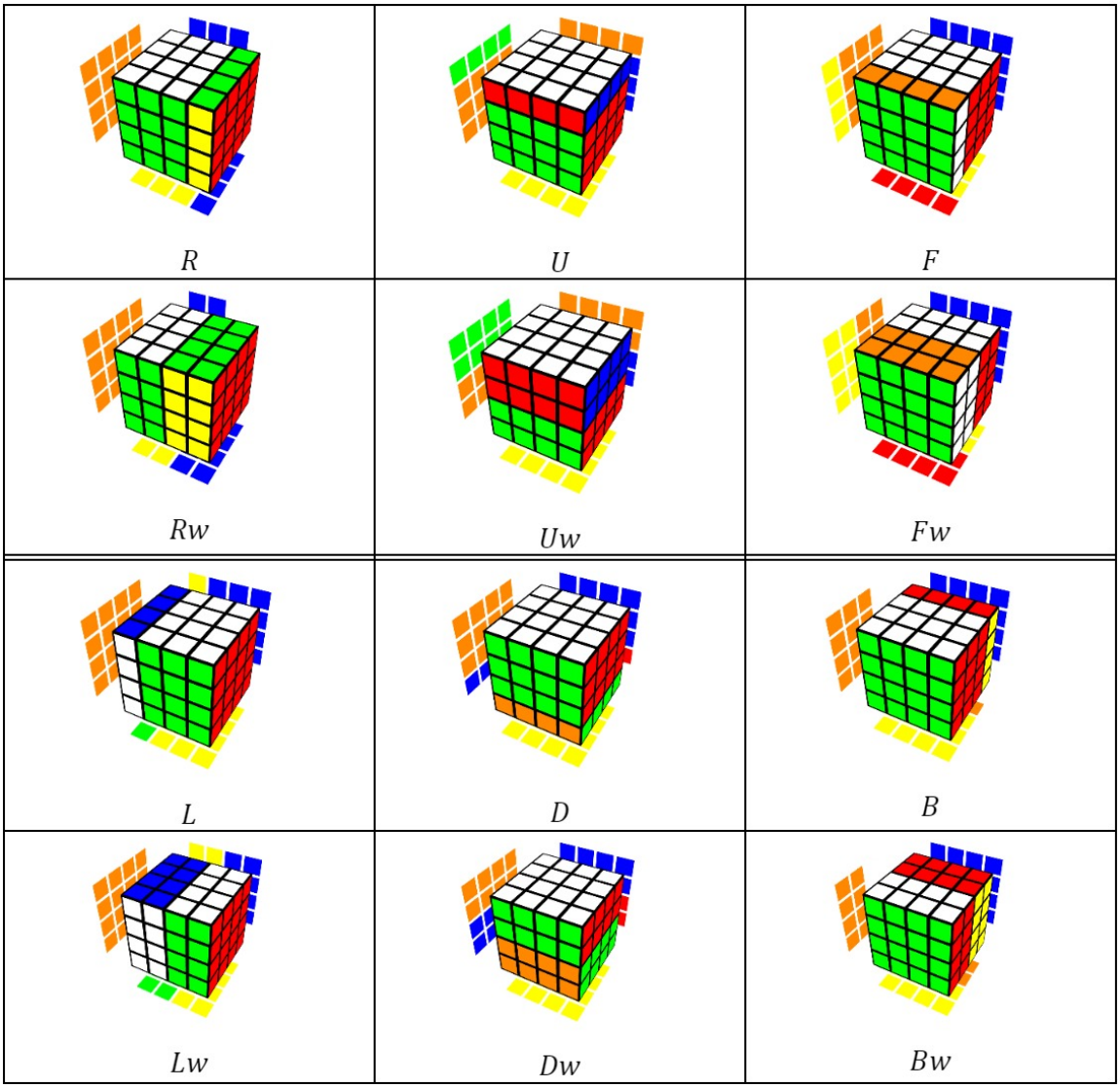


Figure 5: All possible $4 \times 4 \times 4$ moves

These turns behave similarly to those of the $3 \times 3 \times 3$ cube, with the main difference being the w suffix on six. The w represents wide since the closest middle layer is included in the turn.

3 Defining the group

In mathematics, a *group* is a set $\mathbb{G}$ with an operation $*$ that follows the following four axioms [4]:

Closure: for any $a, b \in \mathbb{G}$, $a * b \in \mathbb{G}$;

Associativity: for any $a, b, c \in \mathbb{G}$, $(a * b) * c = a * (b * c)$;

Identity: there exists $e \in \mathbb{G}$ satisfying $e * a = a * e = a$ for all $a \in \mathbb{G}$.
The element e is called the *identity*.

Inverse: for any $a \in \mathbb{G}$, there exists $a^{-1} \in \mathbb{G}$ that satisfies $a * a^{-1} = a^{-1} * a = e$.
The element a^{-1} is called the *inverse* of a.

To each cube can be associated a group of transformations denoted $\mathbb{C}$. We may add a subscript to avoid confusions so that $\mathbb{C}_3$ (resp. $\mathbb{C}_4$) denotes the group of transformations associated to the $3 \times 3 \times 3$ cube (resp. $4 \times 4 \times 4$ cube). For instance, R, R^2DL^2, $RU^2F^{-1}L$ are elements of $\mathbb{C}_3$ and Rw, Rw^2FD, $RU^2Lw^{-1}L$ elements of $\mathbb{C}_4$ (note we have removed the symbol $*$ and simply written RU for $R * U$).

Consequently, the $*$ operation is combining transformations (also called moves) of the cubes. We now check that $\mathbb{C}$ equipped with $*$ is indeed a group by verifying the four axioms of above.

Closure: A move A followed by another move B is also a move, denoted by AB.

Associativity: Given three moves A, B, C, the move A followed by the move BC is the same as the move AB followed by the move C. We write ABC for the composition of these moves.

Identity: The move consisting in leaving all pieces at the same spot is the identity.

Inverse: Given a move A, we may reverse via a move, denoted A^{-1}, so that AA^{-1} is the identity move. For instance, if A is the turn R, then we may consider R^3 and observe that RR^3 is the identity move. Therefore, $A^{-1} = R^{-1} = R^3$.

4 The generator set

It is important to create a set of generators, which are elements that can be used to produce any element in $\mathbb{C}$ [2].

An example of a generating set for an infinite group is the set of integers together with addition; it has a generating set with one element: 1, which is written like so:

$$\{\mathbb{Z}, +\} = \langle 1 \rangle \,.$$

This is because 1 and its additive inverse -1 can create any element in the $\mathbb{Z}$ series. For example,

$$1 + 1 + 1 + 1 = 4\,, \qquad 1 + (-1) = 0\,, \qquad (-1) + (-1) + (-1) = -3\,.$$

The generating sets for the $3 \times 3 \times 3$ and $4 \times 4 \times 4$ cubes contain all the basic turns shown in Figures 3 and 5:

$$\{\mathbb{C}_3, *\} = \langle R, L, U, D, F, B \rangle$$
$$\{\mathbb{C}_4, *\} = \langle R, Rw, L, Lw, U, Uw, D, Dw, F, Fw, B, Bw \rangle \,,$$

where the bracket $\langle , \rangle$ means that the group of transformation is generated by the elements written inside the brackets.

This is because these fundamental moves can be chained together using the $*$ operation to produce move sequences that exist in the cubes' respective $\mathbb{C}$ sets.

5 Permutations and cycles

5.1 Permutations

The different permutations of the $3 \times 3 \times 3$ and $4 \times 4 \times 4$ cubes give the different positions that can be obtained from applying turns to the puzzles, therefore making every element in $\mathbb{C}_3$ and $\mathbb{C}_4$ a permutation possible on the respective puzzle.

More generally, permutations are the ways that objects can be arranged when keeping their order in mind. Take, for example, the objects a, b, c, d [6]:

$$\begin{gathered}(a,b,c,d),(a,b,d,c),(a,c,b,d),(a,c,d,b),(a,d,b,c),(a,d,c,b),\\(b,a,c,d),(b,a,d,c),(b,c,a,d),(b,c,d,a),(b,d,a,c),(b,d,c,a),\\(c,a,b,d),(c,a,d,b),(c,b,a,d),(c,b,d,a),(c,d,a,b),(c,d,b,a),\\(d,a,b,c),(d,a,c,b),(d,b,a,c),(d,b,c,a),(d,c,a,b),(d,c,b,a).\end{gathered}$$

Solving problems with a larger number of objects is sometimes impossible by finding all permutations, as shown above. Instead, it can be useful to simply count the number of permutations, using the permutation formula

$$nPr = \frac{n!}{(n-r)!}$$

where n is the total number of objects and r is the number of objects selected or arranged. In the case above, $n = 4$ and, since no objects are selected, $r = 4$. The total number of permutations is therefore

$$4P4 = \frac{4!}{(4-4)!} = 24\,.$$

5.2 Cycles to reach different permutations of the $3 \times 3 \times 3$ and $4 \times 4 \times 4$

As permutations are the different arrangements of objects, it can be inferred that permutations can be cycled from each other.

Consider permutations (a, b, c, d) and (a, b, d, c) from the previous section. In cycle notation, these are generated by $[c, d]$, as object c moves to where d was, and object d moves to where c was [15].

To visualize this more effectively, the objects being permuted can be considered objects to be placed inside containers, with there being the same number of containers as there are objects. In this example above, the containers can be labeled 1, 2, 3 and 4, making the cycle $[3, 4]$ move object c from container 3 to container 4 and object d from container 4 to container 3.

Additionally, more than two objects can be present in a cycle. For example, to go from (a, b, c, d) to (b, c, d, a), the cycle $[1, 2, 3, 4]$ must be applied. The naming conventions for these cycles depend on the number of elements involved. For example, $[1, 2]$ is a 2-cycle; $[1, 2, 3]$ is a 3-cycle, and so on.

These cycles can be applied to the pieces on the cubes to cycle them around to solve the puzzles. However, to do this, we must create different names for each piece on the $3 \times 3 \times 3$ and $4 \times 4 \times 4$ cube.

5.2.1 Naming the pieces on the $3 \times 3 \times 3$ cube and showing an example cycle

The first step is labeling each piece's face with a name corresponding to the face of the cube, like so [3]:

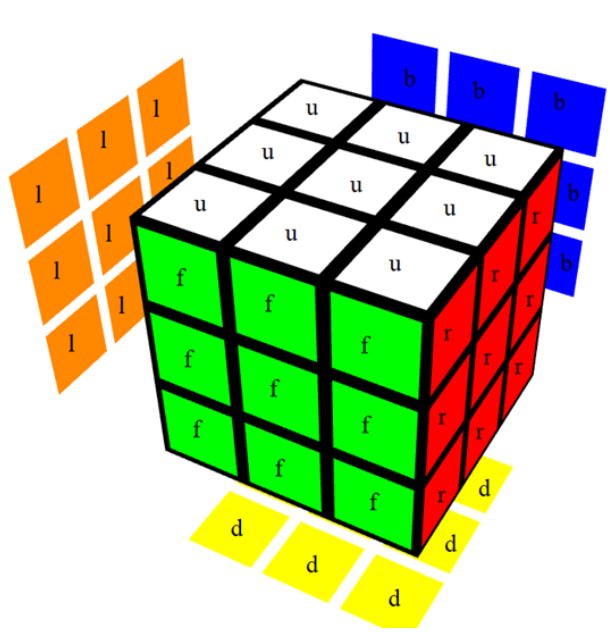


Figure 6: $3 \times 3 \times 3$ piece face labeling

Each piece can be named using this convention. Therefore, centres are signified with one letter, edges with two letters, and corners with three letters. To maintain consistency, the letters for corners should be mentioned in clockwise order. For example, the corner between the front, right, and up faces in Figure 6 is fur, urf or rfu.

The piece cycles for each turn in the generator set can be shown by describing the cycles necessary for a U turn, as all the other turns require analogous cycles on their respective faces.

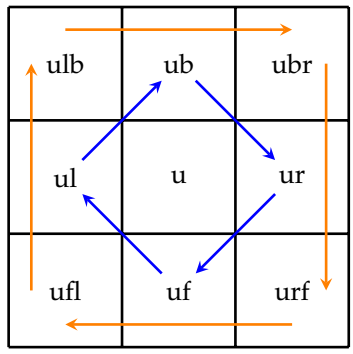


Figure 7: $3 \times 3 \times 3$ *piece cycles from U turn*

From this diagram, we can see that the following piece cycles take place.

$$\begin{aligned} &\text{Edges:} && uf \to ul \to ub \to ur \\ &\text{Corners:} && urf \to ufl \to ulb \to ubr \end{aligned}$$

Thus, using the notation described above, these cycles can be written as follows:

$$[uf, ul, ub, ur][urf, ufl, ulb, ubr]$$

5.2.2 Naming the pieces on the $4 \times 4 \times 4$ cube and showing an example cycle

The same naming scheme can be used to label pieces on the $4 \times 4 \times 4$ cube, like so:

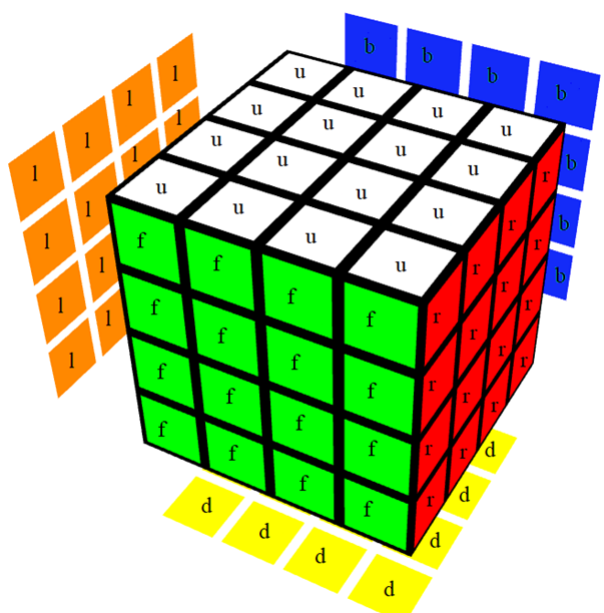


Figure 8: $4 \times 4 \times 4$ *piece face labeling*

However, this naming scheme presents one problem for the $4 \times 4 \times 4$: the repetitions of pieces. While each corner has a unique identity, there are two repeats of each wing and four repeats of each mini-center.

The solution is to give repeated pieces numeric subscripts (1 & 2 in the case of wings and 1, 2, 3 & 4 in the case of mini centres). The wings must be named clockwise for the up and down faces and counter-clockwise for the remaining faces. The mini-centres must be named starting from the top-left centre of each face and naming them in a clockwise direction.

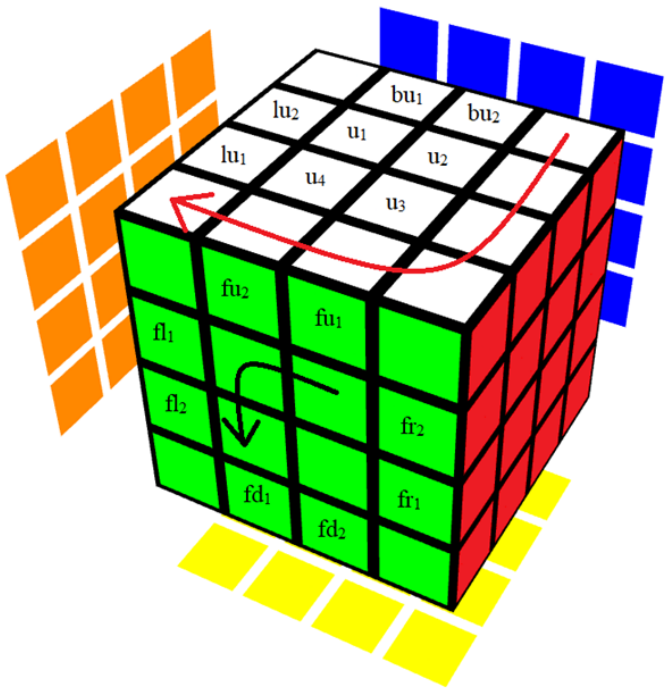


Figure 9: Unique $4 \times 4 \times 4$ piece naming scheme

Consider the following top view of the up face, showing the movement of pieces if a U turn is applied to the cube:

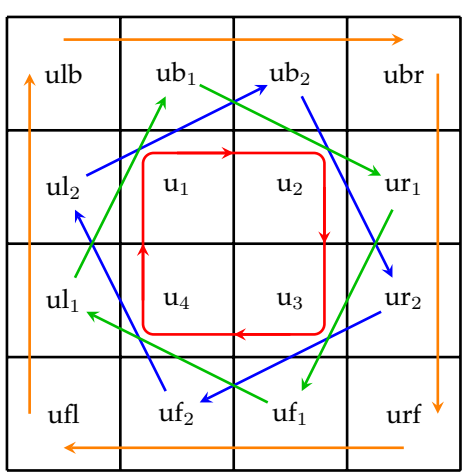


Figure 10: $4 \times 4 \times 4$ piece cycles from U turn

From this diagram, we can see that the following piece movements are taking place:

Mini-centres:	$u_1 \to u_2 \to u_3 \to u_4$
Wings:	$ub_1 \to ur_1 \to uf_1 \to ul_1$
	$ub_2 \to ur_2 \to uf_2 \to ul_2$
Corners:	$ulb \to ubr \to urf \to ufl$

All these cycles can be written together in cycle notation like so:

$$[u_1, u_2, u_3, u_4][ub_1, ur_1, uf_1, ul_1][ub_2, ur_2, uf_2, ul_2][ulb, ubr, urf, ufl]$$

5.3 Decomposing cycles into 2-cycles (and its effect on "illegal cases")

Consider just the centres from the $4 \times 4 \times 4$ cycle mentioned above:

$$[u_1, u_2, u_3, u_4]$$

This 4-cycle can be expressed as a series of 2-cycles, like so [9]:

$$[u_1, u_2]\,[u_1, u_3]\,[u_1, u_4]\,.$$

It can be seen how each object switches with the "container" u_1, a strategy to decompose any cycle into a series of 2-cycles.

generalising this rule, a cycle $[\alpha, \beta, \gamma, \ldots, n]$ with n objects and containers can be decomposed to form the following series of 2-cycles:

$$[\alpha, \beta][\alpha, \gamma] \cdots [\alpha, n]\,.$$

Decomposing cycles helps distinguish between two types of permutations: odd and even [8]. If it is an odd number of 2-cycles, it is an odd permutation, and if it is an even number of cycles, then it is an even permutation. This reintroduces the concept of parity from the beginning of the article, as the permutation types affect the parity both cubes can achieve. Finding the parity both cubes can achieve can help us find configurations to include or rule out for even and odd-layered puzzles.

5.3.1 Decomposing $3 \times 3 \times 3$ cycles

Decomposition can be applied to the previously defined generator set to see the permutations they create. For example, the U move can be split into the following four cycles:

$$[uf, ul, ub, ur][urf, ufl, ulb, ubr] = [uf, ul][uf, ub][uf, ur][urf, ufl][urf, ulb][urf, ubr]$$

which decomposes as 6 different 2-cycles. Thus, the U move is an even permutation. And since the R, L, D, F and B turns are effectively identical cycles for their respective faces, it can be ascertained that the $3 \times 3 \times 3$ cube will always have an even parity. This means that piece cycles that cause an odd parity on the $3 \times 3 \times 3$ are not possible, allowing us to define impossible cases. These will be explained in the next section.

5.3.2 Decomposing $4 \times 4 \times 4$ cycles

As for the $3 \times 3 \times 3$, a U turn can be used to decompose all outer layer turn cycles on the $4 \times 4 \times 4$ cube:

$$\begin{aligned}
&[u_1, u_2, u_3, u_4][ub_1, ur_1, uf_1, ul_1][ub_2, ur_2, uf_2, ul_2][ulb, ubr, urf, ufl] \\
=\ &[u_1, u_2][u_1, u_3][u_1, u_4][ub_1, ur_1][ub_1, uf_1][ub_1, ul_1] \\
&[ub_2, ur_2][ub_2, uf_2][ub_2, ul_2][ulb, ubr][ulb, urf][ulb, ufl]
\end{aligned}$$

which decomposes as 12 different 2-cycles. Thus, the U move is an even permutation. And since the R, L, D, F and B turns are effectively identical cycles for their respective faces, they are even permutations as well.

However, this covers only six of the 12 elements in the generation set, with the other six being the wide moves. The cycles for these wide moves use the cycles from their non-wide variants and extra cycles for the displaced edges and centres in the extra layer being turned. Consider the Uw turn:

$$\begin{aligned}
&[u_1, u_2, u_3, u_4][ub_1, ur_1, uf_1, ul_1][ub_2, ur_2, uf_2, ul_2][ulb, ubr, urf, ufl] \\
&[f_1, l_1, b_1, r_1][f_2, l_2, b_2, r_2][fl_1, bl_2, br_1, fr_2] \\
=\ &[u_1, u_2][u_1, u_3][u_1, u_4][ub_1, ur_1][ub_1, uf_1][ub_1,\ ul_1][ub_2, ur_2][ub_2, uf_2][ub_2, ul_2] \\
&[ulb, ubr][ulb, urf][ulb, ufl][f_1, l_1][f_1, b_1][f_1, r_1][f_2, l_2][f_2, b_2][f_2, r_2] \\
&[fl_1, bl_2][fl_1, br_1][fl_1, fr_2]
\end{aligned}$$

which decomposes as 21 different 2-cycles. This proves that all wide moves (Uw, Dw, Rw, Lw, Fw, Bw) are odd permutations since their cycles can be split into an odd number (21) of 2-cycles.

Since the generator set consists of six turns that result in even parity and six that result in odd parity, no permutations of the $4 \times 4 \times 4$ cube can be ruled out because of even or odd permutations.

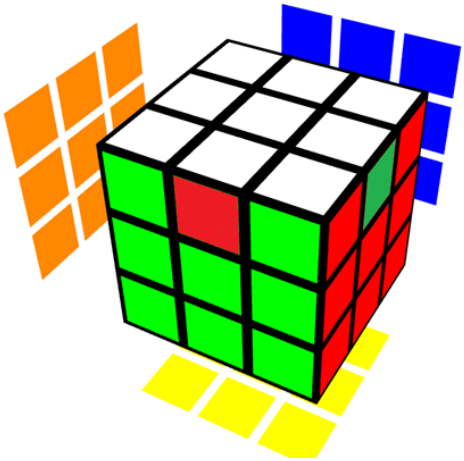
6 Impossible cases

6.1 Impossible $3 \times 3 \times 3$ cases due to permanently-even parity

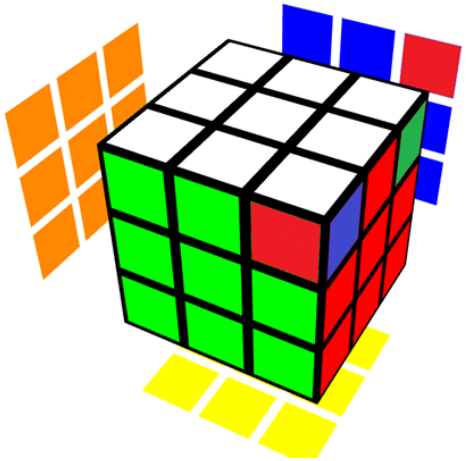
As the $3 \times 3 \times 3$ cube must always have an even parity, all cycles that decompose into an odd number of 2-cycles are impossible.

All possible illegal cases due to odd parity on the $3 \times 3 \times 3$ cube can be simplified to the two base cases shown in Figure 11, given by the following two swaps:

$$\begin{aligned}
&\text{Single Edge Swap:} && [uf, ur] \\
&\text{Single Corner Swap:} && [urf, ubr]
\end{aligned}$$



Edge Case



Corner Case

Figure 11: Impossible odd parity $3 \times 3 \times 3$ cases

Both swaps have one 2-cycle, making them odd permutations that would give the cube odd parity, which is impossible. However, it is essential to note that these cases are impossible individually, as together, the following 2-cycles result in an even parity and are hence possible:

$$[uf, ur][urf, ubr] \,.$$

6.2 Corner twists

Certain types of corner twists can render both the $3 \times 3 \times 3$ and $4 \times 4 \times 4$ cubes unsolvable. Since each corner has three faces, each can be twisted three different ways. Also, since both cubes have the same number and properties of corners, the same illegalities apply to both puzzles. A potentially solvable case appears when the number of corners twisted is even, such as in the following example:

Figure 12: Possible corner twist case

However, there are also examples of an even number of corners twisted that are still unsolvable, like the following example:

Figure 13: Impossible corner twist case

These two examples differ in the twisted corners' rotation direction. For every clockwise twisted corner, there must be a counter-clockwise twisted one to balance it out, regardless of the total number of corners twisted (if it is even).

To accurately predict whether a case of a twisted corner will be solvable or not, we must number the types of corner twists such that the number each corner is designated can be added to find whether the case is solvable:

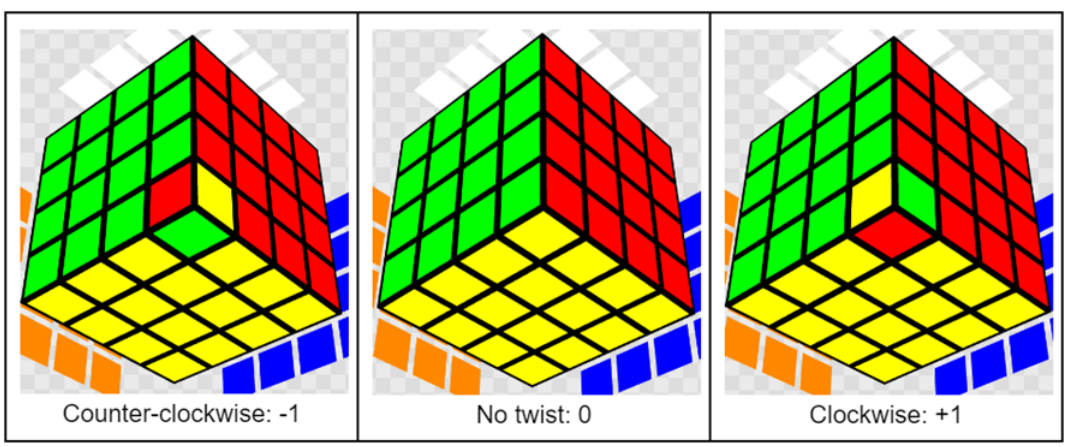


Figure 14: Corner twist numbering

Since we know that each counter-clockwise twist must be balanced out with a clockwise twist for the case to be solvable, we can deduce that the sum must be 0 for the case to be solvable [3].

Consider Figure 12. Two corners are twisted, one twisted clockwise and the other counter-clockwise, making the sum of their corner twist numbers $(-1) + 1 = 0$. Thus, the case is solvable. On the other hand, the cube in Figure 13 is unsolvable since both corners are twisted counter-clockwise, giving the sum $(-1) + (-1) = -2 \neq 0$.

6.3 Edge flips on the $3 \times 3 \times 3$

Like corner twists, certain types of edge flips render the cube unsolvable. This time, however, there is no rotational direction like with the corners, meaning that an edge can be denoted in only the following two ways [3]:

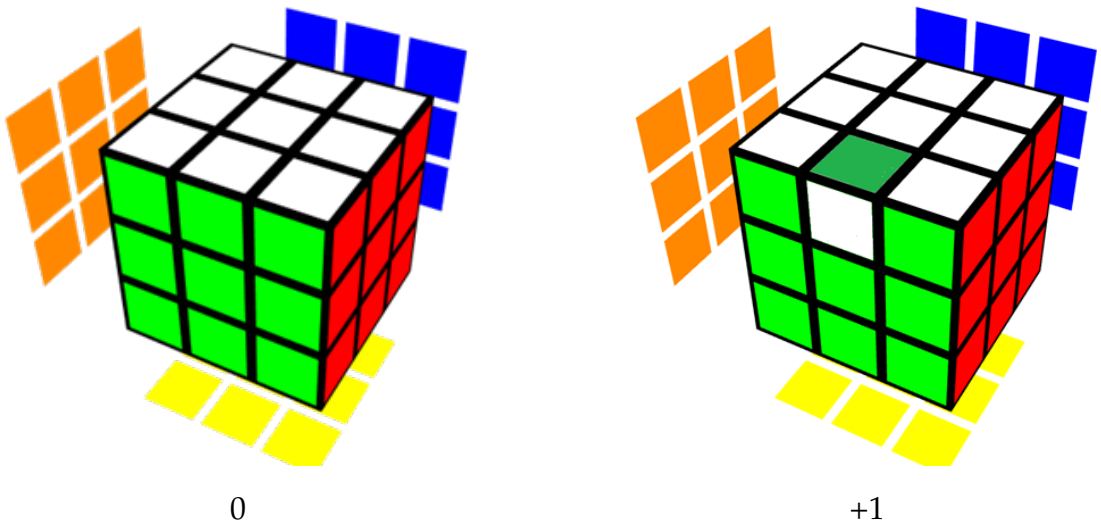


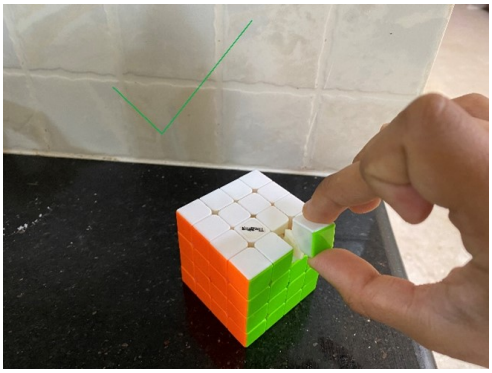
Figure 15: Edge flip numbering

For a configuration to be possible, the number of flipped edges must be even. This means that, while the flipped edge case in Figure 15 is impossible (1 is odd), the following case is possible:

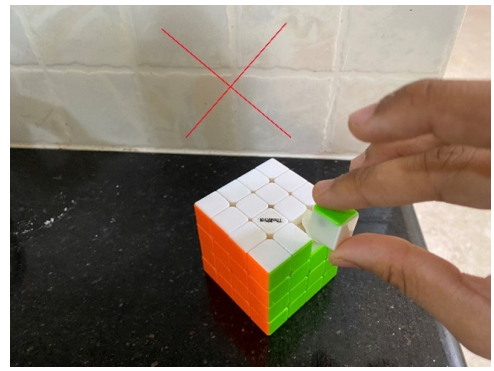
Figure 16: Possible $3 \times 3 \times 3$ edge flip case

It is important to note that wing pieces on the $4 \times 4 \times 4$ cube cannot be physically flipped, giving each wing piece one possible orientation and, therefore, no impossible cases in this regard. The reason for this is their unique structure [11]:

Figure 17: $4 \times 4 \times 4$ *wing piece*



The correct way



The incorrect way

Figure 18: Correct and incorrect way of orienting wing piece

7 The number of $3 \times 3 \times 3$ Rubik's Cube configurations

7.1 Corners

The $3 \times 3 \times 3$ cube has 8 corner slots and 8 corner pieces that can fit into them in any configuration, of which there are $8!$.

As mentioned in the corner twisting section, for an even number of corners twisted, the twist direction determines whether the cube is solvable. Consider the last two corners being twisted such that the cube is in a legal configuration. If one corner is fixed to a specific state (-1, 0, 1), then the other would only have one possible state, the opposite of the fixed corner (1, 0, or -1, respectively). This means that the total configurations must be divided by 3 since the last corner piece's twist direction is predetermined. As a result, the total number of configurations of the corner pieces is

$$8! \times \frac{3^8}{3} = 8! \times 3^7 \,.$$

7.2 Edges

The $3 \times 3 \times 3$ cube has 12 edge slots and 12 edge pieces that can fit into them in any configuration, of which there are $12!$. However, because of the $3 \times 3 \times 3$'s permanently-even parity, the positions of the last two edges are predetermined by the corner positioning to ensure that the even parity is maintained, therefore ruling out half of the cases where parity is odd, meaning that the final value must be divided by 2.

When considering the flip of the edges, the numbering scheme developed in the edge flips section must be used. The flip of the last edge is predetermined to make the total number of flipped edges even/divisible by 2. This means that the total configurations of edge pieces can be divided by 2 as there are two possible ways an edge can be positioned (0 or +1), therefore making the total number of configurations of the edges

$$\frac{12!}{2} \times \frac{2^{12}}{2} = 12! \times 2^{10}\,.$$

7.3 The final calculation

Multiplying the corner and edge calculations, we find that the total number of legal configurations of the $3 \times 3 \times 3$ Rubik's Cube are

$$8! \times 3^7 \times 12! \times 2^{10} = 4.325 \times 10^{19}\,.$$

8 The number of $4 \times 4 \times 4$ Rubik's Cube configurations

8.1 Corners

The corners of the $4 \times 4 \times 4$ cube act like the $3 \times 3 \times 3$ cube. Therefore, their total number of configurations is

$$8! \times \frac{3^8}{3} = 8! \times 3^7\,.$$

8.2 Mini centres

The $4 \times 4 \times 4$ cube has 24 mini centres, 4 per face. Unlike edges and corners, each mini-centre has only one colour, making its orientation irrelevant. Thus, we only have to focus on their position.

While there are 24 mini centres, there are only 6 distinct ones, each with three indistinguishable copies. Therefore, using the permutation formula for nPr given earlier, the total number of centre configurations is

$$\frac{24!}{(4!)^6}\,.$$

The $4!$ is raised to the power of 6 due to the 4 identical centres on each of the 6 faces.

8.3 Wings

Due to wing pieces being such that there are no illegal cases on the $4 \times 4 \times 4$ cube, the total number of configurations of the wings is

$$24!\,.$$

8.4 Rotational equality

Due to its lack of fixed centres, the angles at which the entire $4 \times 4 \times 4$ cube can be viewed must be considered. Without doing this, each viewing angle would be considered a different permutation, causing the total number of configurations to be overcounted.

Each face can be viewed from four angles. Multiplying by 6 for all faces gets $4 \times 6 = 24$, which the configurations calculation should be divided by.

8.5 The final calculation

Using the information above, the total number of legal configurations of the $4 \times 4 \times 4$ Rubik's Cube is

$$\frac{8! \times 3^7 \times \frac{24!}{(4!)^6} \times 24!}{24} = 7.401 \times 10^{45}\,.$$

9 Generalising for the $N \times N \times N$ Rubik's Cube

9.1 Corners

Like with the $3 \times 3 \times 3$ and $4 \times 4 \times 4$ cubes, the corner pieces of an $N \times N \times N$ Rubik's Cube will always have the same number of configurations since the number of corners is constant:

$$8! \times \frac{3^8}{3} = 8! \times 3^7\,.$$

9.2 Centres

Like the $4 \times 4 \times 4$ cube, as the value of N increases, so does the number of distinct centre pieces that can only be cycled between each other (excluding the fixed centre if there is one, as it cannot be cycled). Consider the $5 \times 5 \times 5$ cube:

	type 1	type 2	type 1	
	type 2		type 2	
	type 1	type 2	type 1	

Figure 19: Distinct types of centres

For each distinct centre type, the centre calculation used for the $4 \times 4 \times 4$ must be repeated. Therefore, to generalise, we must create a relationship between N and the number of different centre types.

N	2	3	4	5	6	7	8	9	10	11
D	0	0	1	2	4	6	9	12	16	20

Table 1: N vs. number (D) of distinct centre types

While it seems that there is no direct relationship, consider only the even values of N.

N	2	4	6	8	10
D	0	1	4	9	16

Table 2: N (even) vs. number (D) of distinct centre types

This forms the square numbers sequence, which can be used to create the following relationship:

$$D = \left(\frac{N-2}{2}\right)^2 .$$

However, we have only established its use for even values of N. We must now test it against odd values of N.

$$\begin{aligned} D(3) &= 0.25 \\ D(5) &= 2.25 \\ D(7) &= 6.25 \\ D(9) &= 12.25 \\ D(11) &= 20.25 \,. \end{aligned}$$

As can be seen, the integer part of these results matches the values in the table above. However, their decimal part (0.25) makes them useless for calculating the total configurations. To solve this, we must use the floor function to round these values down when N is odd. Thus, the generalised total number of centre configurations is

$$\left(\frac{24!}{(4!)^6}\right)^{\left\lfloor \left(\frac{N-2}{2}\right)^2 \right\rfloor} .$$

9.3 $3 \times 3 \times 3$-style edges

All odd-layered puzzles have one centre edge that behaves like a $3 \times 3 \times 3$ edge, meaning the same calculation must be applied. For example, on the $5 \times 5 \times 5$ cube, it would be the following edges:

		*		
*				*
		*		

Figure 20: $3 \times 3 \times 3$-style edges

To ensure this calculation is only applied to odd-layered cubes, the expression must be raised to the power N mod 2, which is the remainder when N is divided by 2: 0 when N is even and 1 when N is odd). Therefore, the final expression for the total number of $3 \times 3 \times 3$-style edge configurations is:

$$\left(12! \times 2^{10}\right)^{N \bmod 2} .$$

9.4 $4 \times 4 \times 4$-style edges

Like the centres, as N increases, so does the number of distinct edge types that cycle between each other (excluding the middle, $3 \times 3 \times 3$-style centre). Consider the $6 \times 6 \times 6$ cube:

					type 1
					type 2
					type 2
					type 1

Figure 21: Distinct types of edges

The $4 \times 4 \times 4$ edge calculation must be repeated for each distinct edge type. Therefore, to generalise, we must create a relationship between N and the number of different centre types.

N	2	3	4	5	6	7	8	9	10	11
E	0	0	1	1	2	2	3	3	4	4

Table 3: N vs. number (E) of distinct edge types

Yet again, a relationship can be formed between E and the even values of N:

N	2	4	6	8	10
E	0	1	2	3	4

Table 4: N (even) vs. number (E) of distinct edge types

This relationship is

$$E = \frac{N-2}{2}\,.$$

When testing it against odd values of N, we find a similar relationship to that of the centres:

$$\begin{aligned} E(3) &= 0.5 \\ E(5) &= 1.5 \\ E(7) &= 2.5 \\ E(9) &= 3.5 \\ E(11) &= 4.5\,. \end{aligned}$$

Once again, the floor function can be used to remove the decimal part (0.5) of this otherwise correct integer relationship, forming the following expression for the total configurations of $4 \times 4 \times 4$-style edges:

$$(24!)^{\left\lfloor \frac{N-2}{2} \right\rfloor}\,.$$

9.5 Rotational equality

As mentioned in the $4 \times 4 \times 4$ section, dividing by 24 to achieve rotational equality must only be done on even-layered puzzles. For the generalisation, the 24 must be raised to the power $(N+1) \bmod 2$. If N is even, then $(N+1) \bmod 2 = 1$, applying rotational equality, but if N is odd, then $(N+1) \bmod 2 = 0$, removing rotational equality.

9.6 The final generalisation

Using the individual calculations shown above, the total number of configurations of an $N \times N \times N$ Rubik's Cube is

$$\frac{3^7 \times 8! \times \left(\frac{24!}{(4!)^6}\right)^{\left\lfloor \left(\frac{N-2}{2}\right)^2 \right\rfloor} \times (12! \times 2^{10})^{N \bmod 2} \times (24!)^{\left\lfloor \frac{N-2}{2} \right\rfloor}}{24^{(N+1) \bmod 2}}.$$

9.7 Verifying the generalisation

9.7.1 $3 \times 3 \times 3$ cube

The total number of configurations is

$$\frac{3^7 \times 8! \times \left(\frac{24!}{(4!)^6}\right)^{\left\lfloor \left(\frac{3-2}{2}\right)^2 \right\rfloor} \times (12! \times 2^{10})^{3 \bmod 2} \times (24!)^{\left\lfloor \frac{3-2}{2} \right\rfloor}}{24^{4 \bmod 2}} = 3^7 \times 8! \times 12! \times 2^{10}$$

which is correct.

9.7.2 $4 \times 4 \times 4$ cube

The total number of configurations is

$$\frac{3^7 \times 8! \times \left(\frac{24!}{(4!)^6}\right)^{\left\lfloor \left(\frac{4-2}{2}\right)^2 \right\rfloor} \times (12! \times 2^{10})^{4 \bmod 2} \times (24!)^{\left\lfloor \frac{4-2}{2} \right\rfloor}}{24^{5 \bmod 2}} = \frac{3^7 \times 8! \times \frac{24!}{(4!)^6} \times 24!}{24}$$

which is correct.

9.7.3 $5 \times 5 \times 5$ cube

The total number of configurations is

$$\frac{3^7 \times 8! \times \left(\frac{24!}{(4!)^6}\right)^{\left\lfloor \left(\frac{5-2}{2}\right)^2 \right\rfloor} \times (12! \times 2^{10})^{5 \bmod 2} \times (24!)^{\left\lfloor \frac{5-2}{2} \right\rfloor}}{24^{6 \bmod 2}} = 2.83 \times 10^{74}$$

which is correct according to The Rubik Zone [13].

9.7.4 $6 \times 6 \times 6$ cube

The total number of configurations is

$$\frac{3^7 \times 8! \times \left(\frac{24!}{(4!)^6}\right)^{\left\lfloor \left(\frac{6-2}{2}\right)^2 \right\rfloor} \times (12! \times 2^{10})^{6 \bmod 2} \times (24!)^{\left\lfloor \frac{6-2}{2} \right\rfloor}}{24^{7 \bmod 2}} = 1.57 \times 10^{116}$$

which is correct according to The Rubik Zone [13].

10 Conclusion

This article aimed to find an expression for the total number of possible configurations of an $N \times N \times N$ Rubik's Cube. The methodology was to find the total configurations of the odd-layered $3 \times 3 \times 3$ and even-layered $4 \times 4 \times 4$ and to use their calculation properties to derive a general expression for the $N \times N \times N$ Rubik's Cube.

Firstly, the $3 \times 3 \times 3$ and $4 \times 4 \times 4$ cubes were defined as groups by following the four axioms, after which their generator sets were produced. Then, the parities produced by these generator sets were examined to find that odd-layered cubes have permanently-even parity, while even-layered cubes can have either parity. This fact was used to find the impossible cases on both cubes.

Using the knowledge gained about piece cycles and impossible cases, the total configurations of the $3 \times 3 \times 3$ and $4 \times 4 \times 4$ cubes were calculated. Based on these calculations, the generalisation was completed, utilizing the modulus and floor functions to allow the generalisation to work when N is even or odd.

In conducting this investigation, I learned of the $N \times N \times N$ generalisations of Chris Hardwick [5], Christopher Mowla [10], and Herbert Kociemba [7], all of whom used different methodologies like trigonometry, primes, and recursion to create their formulae. An extension of this study could be evaluating how they defined the cube differently to utilize other concepts in their generalisations.

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